

Scenario 2: Standalone Local Mode (Webcam + YOLOv8)

Robotics & Edge AI Workshop — Offline Simulation & Computer Vision Demonstration Guide

1. Computer Vision & YOLO Theory

Core Concept: When physical robot hardware is absent or unavailable, vision algorithms can be demonstrated directly on a laptop using the integrated or USB webcam and a local deep learning model.

Key Concepts to Present:

- **Single-Stage Object Detection (YOLO - You Only Look Once):** Unlike legacy two-stage detectors (e.g., R-CNN) that first generate region proposals and then classify them, YOLO reframes detection as a single regression problem. A single convolutional/transformer neural network divides the image into a grid and predicts bounding boxes, confidence scores, and class labels simultaneously in a single forward pass.
- **Inference Pipeline:** Frame Capture (OpenCV) → Tensor Resizing/Normalization → Neural Network Forward Pass → Non-Maximum Suppression (NMS) → Bounding Box Rendering.
- **YOLOv8 Nano (yolov8n):** Designed specifically for real-time edge processing and CPU inference (~3.2 million parameters), achieving 30+ FPS even without dedicated GPU.

Metric	YOLOv8 Nano (yolov8n)	Full YOLOv8 Extra (yolov8x)
Parameters	~3.2 Million	~68.2 Million
Inference Speed	Real-time (CPU friendly)	Requires High-end GPU
Primary Use Case	Robotics edge devices & laptops	Offline server batch processing

2. Installation & Setup

```
# Install Ultralytics YOLO and computer vision dependencies
pip install ultralytics opencv-python numpy
```

3. Execution Commands

```
# Run the standalone webcam YOLO script
python webcam_vision.py

# Controls:
# Press 'q' inside the video window to quit.
```

4. Common Issues & Fixes in Workshop

- **'Camera index out of range' or OpenCV backend error:**
 - Windows uses DirectShow (cv2.CAP_DSHOW) or MSMF for webcams. The script tests both automatically.
 - **Windows Privacy:** Open **Windows Settings** → **Privacy & security** → **Camera**. Ensure *'Let desktop apps access your camera'* is **ON**.
 - Ensure no background apps (Zoom, Teams, Discord, Browser) are currently holding a lock on the webcam.
- **First Run Delay:** On first execution, Ultralytics downloads yolov8n.pt (~6 MB) automatically from GitHub.